

Chapter 8

Measures for distances and mixedness

Theorem 8.1 (Russo-Dye) ...

8.1 Norms

A *matrix norm* is a vector norm on the space of matrices $\mathcal{M}_{n,m}(\mathbb{C})$. That is, it is a positive functional which is convex (i.e., it satisfies the triangle inequality $\|A+B\| \leq \|A\| + \|B\|$ for all A, B) and such that $\|aA\| = |a| \|A\|$ for all $a \in \mathbb{C}$ and ($\|A\| = 0 \Rightarrow A = 0$). For square matrices some norms have the additional property of being *sub-multiplicative*, meaning that $\|AB\| \leq \|A\| \|B\|$ holds. Sometimes this property is included in the definition of matrix norms. Equipped with a sub-multiplicative norm, $\mathcal{M}_d(\mathbb{C})$ becomes a *Banach algebra* (see Sec.1.6).

An important class of sub-multiplicative matrix norms on $\mathcal{M}_d(\mathbb{C})$ are those which are *unitarily invariant* in the sense that $\|UAV\| = \|A\|$ for all unitaries U and V . Since every element $A \in \mathcal{M}_d(\mathbb{C})$ has a singular value decomposition $A = U \text{diag}(s_1, \dots, s_n) V$, a unitarily invariant norm depends solely on the *singular values* $\{s_i(A)\}$.¹ The two mostly used classes of unitarily invariant norms are:

- *Schatten p -norms*: $\|A\|_p := \left(\sum_{i=1}^d [s_i(A)]^p \right)^{1/p}$ for any $p \geq 1$,
- *Ky Fan k -norms*: $\|A\|_{(k)} := \sum_{i=1}^k s_i(A)$ where $k = 1, \dots, d$ and the singular values are decreasingly ordered.

Both classes have a monotonicity property in their parameter. Evidently, $\|A\|_{(k)} \geq$

¹In fact, unitarily invariant norms on $\mathcal{M}_d$ are in one-to-one correspondence with norms $f : \mathbb{R}^d \rightarrow \mathbb{R}^+$ which are (i) permutational invariant and (ii) depend only on absolute values. Such a vector norm f is called *symmetric gauge function* and it give rise to a unitarily invariant norm on $\mathcal{M}_d \ni A$ via $\|A\| = f([s_1(A), \dots, s_d(A)])$.

$\|A\|_{(k')}$ for $k \geq k'$, but also

$$\|A\|_1 \geq \|A\|_p \geq \|A\|_{p'} \geq \|A\|_\infty, \quad (8.1)$$

for all $1 \leq p \leq p'$. Moreover, $p < p'$ leads to a strict inequality iff A has rank at least two.² The norm $\|A\|_\infty = \|A\|_{(1)}$ is called the *operator norm* and it can be defined alternatively as $\|A\|_\infty = \sup_{\psi, \phi} |\langle \psi | A | \phi \rangle|$ with supremum over all unit vectors $\phi, \psi \in \mathbb{C}^d$. With the operator norm $\mathcal{M}_d(\mathbb{C})$ becomes a C^* -algebra as the C^* -equality $\|A^\dagger A\| = \|A\|^2$ is satisfied (see Sec.1.6). This property uniquely identifies the operator norm. The norm $\|A\|_1 = \|A\|_{(d)} = \text{tr} [|A|]$ is called the *trace norm*. Trace norm and operator norm are the largest/smallest norms among all unitarily invariant norms in the sense that $\|A\|_\infty \leq \|A\| / \|\rho\| \leq \|A\|_1$, where $\rho = |1\rangle\langle 1|$ takes care of normalization. The norm $\|A\|_2$ is called *Hilbert-Schmidt norm* or *Frobenius norm*. It can be regarded as arising from the Hilbert-Schmidt scalar product (see Sec.2.3) and it coincides with the l_2 -norm. The latter is defined within the class of l_p norms $\|A\|_{l_p} := (\sum_{kl} |A_{kl}|^p)^{1/p}$ for $p \geq 1$.³ They are bounded by the Schatten p -norms (which one may call L_p -norms in this context) via $\|A\|_{l_p} \geq \|A\|_p$ for $p \in [1, 2]$ and with inequality in the opposite direction for $p \geq 2$. Apart from l_2 the l_p -norms are not unitarily invariant.

Besides being sub-multiplicative unitarily invariant norms exhibit some very useful properties which are collected in the following:

Theorem 8.2 (Unitarily invariant norms) *Let $\|\cdot\|$ be any unitarily invariant norm on $\mathcal{M}_d(\mathbb{C})$. Then the following holds for $A, B \in \mathcal{M}_d(\mathbb{C})$:*

1. Hölder's inequality for $p \geq 1$ and $p^{-1} + q^{-1} = 1$:

$$\|AB\| \leq \| |A|^p \|^{1/p} \| |B|^q \|^{1/q}. \quad (8.2)$$

2. Lidskii's inequalities: *If A and B are Hermitian and $E^\downarrow(A)$, $E^\uparrow(A)$ denote the diagonal matrices with decreasingly/increasingly ordered eigenvalues of A respectively, then⁴*

$$\|A - B\| \begin{cases} \leq \|E^\downarrow(A) - E^\uparrow(B)\| \\ \geq \|E^\downarrow(A) - E^\downarrow(B)\| \end{cases}. \quad (8.3)$$

3. *If AB is normal, then $\|AB\| \leq \|BA\|$,*

4. *For every non-negative concave function f on $[0, \infty)$ and all $A, B \geq 0$:*

$$\|f(A) + f(B)\| \geq \|f(A + B)\|. \quad (8.4)$$

The inequality is reversed if f is non-negative and convex on $[0, \infty)$ with $f(0) = 0$.

²In infinite dimensions the analogue of Eq.(8.1), in particular $\|A\|_1 \geq \|A\|_2 \geq \|A\|_\infty$ gives rise to the inclusion *trace class operators* $\subset$ *Hilbert-Schmidt class operators* $\subset$ *bounded operators*.

³Note that in the definition of the l_p -norms as well as for the Schatten p -norms $p \geq 1$ is crucial for convexity.

⁴If one drops the restriction on A and B to be Hermitian and replaces eigenvalues by singular values in the diagonal matrices, then still the lower bound holds true.

5. Ky Fan dominance: if we have $\|A\|_{(k)} \geq \|B\|_{(k)}$ for all Ky-Fan norms ($k = 1, \dots, d$), then $\|A\| \geq \|B\|$ for every unitarily invariant norm.

6. Araki-Lieb-Thirring inequalities: If $A, B \geq 0$ and $p \geq 1$, then

$$\|(BAB)^p\| \leq \|B^p A^p B^p\|, \quad \| |BA|^p \| \leq \|B^p A^p\|, \quad (8.5)$$

and the reversed inequalities hold for $p \in (0, 1]$.

Applying Hölder's inequality to the trace norm and using that $|\operatorname{tr}[A^\dagger B]| \leq \operatorname{tr}[|A^\dagger B|]$ leads to a useful Cauchy-Schwarz type version of the inequality, namely:

$$|\operatorname{tr}[A^\dagger B]| \leq \|A\|_p \|B\|_q. \quad (8.6)$$

For $p = 2$ this is indeed the Chauchy-Schwarz inequality for the Hilbert-Schmidt inner product. Another simple application of Hölder's inequality is a relation between different p -norms opposite to the direction of Eq.(8.1): setting $B = \mathbb{1}$ in Eq.(8.2) and applying it to the p -norm we arrive at

$$\|A\|_p \leq d^{\frac{1}{p} - \frac{1}{p'}} \|A\|_{p'}, \quad \text{for } p' \geq p \geq 1. \quad (8.7)$$

In a similar vein we can derive the following moment inequality which generalizes the well known variance inequality $\langle X^2 \rangle \geq \langle X \rangle^2$ known for Hermitian operators:

Proposition 8.1 (Moment inequalities) Consider $1 \leq p \leq p'$ and $\rho, X \in \mathcal{M}_d(\mathbb{C})$ where ρ is a density operator and $X \geq 0$. Then

$$\operatorname{tr}[\rho X^p]^{p'} \leq \operatorname{tr}[\rho X^{p'}]^p. \quad (8.8)$$

PROOF This follows from Hölder's inequality $\|AB\|_1 \leq \|A\|_{\tilde{p}} \|B\|_{\tilde{q}}$ by setting $\tilde{q} = p'/p$, $A = \rho^{1/\tilde{p}}$ and $B = \rho^{\frac{p}{2p'}} X^p \rho^{\frac{p}{2p'}}$. $\square$

Variational characterization of norms: In the same spirit as the extremal eigenvalue of a Hermitian matrix can be obtained variationally, we can characterize the Schatten p -norms and the Ky-Fan norms as the solution of optimization problems. For the latter it holds for instance that

$$\|A\|_{(k)} = \sup \{ |\operatorname{tr}[A^\dagger B]| \mid B^\dagger B \leq \mathbb{1}, \operatorname{rank}(B) \leq k \}. \quad (8.9)$$

Inequality (8.6) is the basis for an analogous variational characterization of the Schatten p -norms:

Theorem 8.3 (Variational ways to p -norms) Let $A, P \in \mathcal{M}_d(\mathbb{C})$ with $P \geq 0$ and $p^{-1} + q^{-1} = 1$ with $p \geq 1$. Then

$$\|A\|_p = \sup \{ |\operatorname{tr}[A^\dagger B]| \mid \|B\|_q = 1 \}, \quad (8.10)$$

$$\|A\|_1 = \sup \{ |\operatorname{tr}[A^\dagger U]| \mid UU^\dagger = \mathbb{1} \}, \quad (8.11)$$

$$\|A\|_\infty = \sup \{ |\langle \psi | A | \phi \rangle| \mid \|\psi\| = \|\phi\| = 1 \}, \quad (8.12)$$

$$\|P\|_p^p = \frac{1}{p} \sup_{X \geq 0} \left[\operatorname{tr}[PX] - \frac{1}{q} \|X\|_q^q \right]. \quad (8.13)$$

PROOF In order to arrive at Eq.(8.10) first note that the supremum is an upper bound by Eq.(8.6). The supremum is attained if we set $B = UP^{p-1}\text{tr}[P^p]^{-1/q}$ constructed from the polar decomposition of $A = UP$. Eqs.(8.11,8.12) are special cases which arise from (8.10) due to the fact that the set of operators with $\|B\|_q \leq 1$ is convex and, since we maximize a convex functional, the maximum is attained at an extreme point. The extreme points are the set of unitaries for $q = 1$ and normalized rank-one operators for $q = \infty$, respectively.

... .. $\square$

Eq.(8.10) is the reason why p -norms and q -norms are called *dual* to each other if $p^{-1} + q^{-1} = 1$. In fact, for every norm $\|\cdot\|$ on $\mathcal{M}_{n,m}$ we can define a *dual norm* $\|\cdot\|^D$ w.r.t. the Hilbert Schmidt inner product via

$$\|A\|^D := \sup\{|\text{tr}[A^\dagger B]| \mid B \in \mathcal{M}_{n,m}(\mathbb{C}), \|B\| = 1\}. \quad (8.14)$$

The dual norm is unitarily invariant iff $\|\cdot\|$ is.

8.2 Entropies

8.2.1 Information theoretic origin

8.2.2 Mathematical origin

axiomatization ..

$$S(\rho) = -\frac{\partial}{\partial p} \text{tr}[\rho^p] \Big|_{p=1}. \quad (8.15)$$

8.2.3 Physical origin

Boltzmann, time asymmetry & co., relation to the above, statistical interpretation

8.3 Majorization

Majorization is a stronger means of saying that a probability distribution/density operator is more mixed than another one. While entropies and the respective norms provide an ordering in the space of probability distributions/density operators, majorization leaves us with a partial order. We will, in fact, see that one distribution is more mixed than another in the majorization sense iff this holds true for all entropy-type functionals.

Consider two vectors $x, y \in \mathbb{R}^d$. We say that x is majorized by y , and write $x \prec y$, iff for the sum of decreasingly ordered components we have

$$\sum_{i=1}^k x_i^\downarrow \leq \sum_{i=1}^k y_i^\downarrow, \quad \forall k = 1, \dots, d \text{ with equality for } k = d. \quad (8.16)$$

Note that if x describes any discrete probability distribution, then

$$\left(\frac{1}{d}, \dots, \frac{1}{d}\right) \prec x \prec (1, 0, \dots, 0), \quad (8.17)$$

consistent with our intuition about what is the ‘maximally mixed’ distribution and which is a most ‘pure’ distribution. If only the d inequalities in Eq.(8.16) are satisfied but equality for $k = d$ is not required, then we say that x is *weakly submajorized* by y and we write $x \prec_w y$.

The basic results in the context of majorization are:

Theorem 8.4 (Majorization) *Let $x, y \in \mathbb{R}^d$. The following statements are equivalent:*

1. x is majorized by y ,
2. $x = My$ for some doubly stochastic⁵ matrix $M \in \mathcal{M}_d(\mathbb{R})$,
3. $x = My$ where $M_{kl} = O_{kl}^2$ for some real orthogonal matrix⁶ $O \in \mathcal{M}_d(\mathbb{R})$,
4. $\sum_i f(x_i) \leq \sum_i f(y_i)$ for all convex functions $f : \mathbb{R} \rightarrow \mathbb{R}$.

Theorem 8.5 (Submajorization) *Let $x, y \in \mathbb{R}^d$. The following statements are equivalent:*

1. $x \prec_w y$
2. $x = My$ for some doubly substochastic⁷ matrix $M \in \mathcal{M}_d(\mathbb{R})$,
3. There exists a vector $z \in \mathbb{R}^d$ such that $z \prec y$ and $x_i \leq z_i$ for all i ,
4. $\sum_i f(x_i) \leq \sum_i f(y_i)$ for all monotonically increasing convex functions $f : \mathbb{R} \rightarrow \mathbb{R}$.

Theorem 8.6 (Birkhoff) *The set of doubly stochastic matrices in $\mathcal{M}_d(\mathbb{R})$ is a convex polytope. Its extreme points are the permutation matrices. Similarly, the set of doubly substochastic matrices is a convex polytope. Its extreme points are all matrices which have at most one entry 1 in each column and each row and all other entries zero.*

Excursion 8.1 (Diagonals, eigenvalues, singular values) *Consider a Hermitian matrix $H \in \mathcal{M}_d(\mathbb{C})$ with eigenvalues $\{\lambda_i\}$ and diagonal elements $D_i := H_{ii}$. Using that we can diagonalize H via a unitary U , we get the relation $D_i = \sum_j |U_{ij}|^2 \lambda_j$ and since $[|U_{ij}|^2]$ is doubly stochastic we have*

$$\lambda \succ D. \quad (8.18)$$

⁵Doubly stochastic means that M is non-negative ($M_{kl} \geq 0$) and $\sum_k M_{kl} = \sum_j M_{ij} = 1$ for all l, i .

⁶If $U \in \mathcal{M}_d(\mathbb{C})$ is any unitary, then the matrix with components $M_{kl} = |U_{kl}|^2$ is doubly stochastic and called a *unistochastic* matrix. If U real orthogonal, then M is called *orthostochastic*.

⁷Doubly substochastic means that M is non-negative and all columns and rows have a sum which is at most 1.

Given vectors $D, \lambda \in \mathbb{R}^d$ Eq.(8.18) is thus a necessary condition for the existence of a Hermitian matrix with eigenvalues λ and diagonal entries D . As shown by Horn it is also a sufficient one. If $\lambda, D \in \mathbb{C}^d$ are complex, then Mirsky showed that there exists a complex matrix in $\mathcal{M}_d$ with eigenvalues λ_i and diagonal entries D_i iff $\sum_i \lambda_i = \sum_i D_i$, i.e., the trace remains the only relevant condition. In a similar vein we can ask for the relation between eigenvalues and singular values and obtain again a majorization like condition: let $s \in \mathbb{R}_+^d$ and $\lambda \in \mathbb{C}^d$, then there exists a matrix with singular values s_i and eigenvalues λ_i iff

$$\prod_{i=1}^k s_i \geq \left| \prod_{i=1}^k \lambda_i \right| \quad \forall k = 1, \dots, d \quad \text{with equality for } k = d, \quad (8.19)$$

where both s and $|\lambda|$ are arranged in decreasing order. That is, if there is no zero element we can express this as $\log s \succ \log |\lambda|$ which, in particular implies $|\lambda| \prec_w s$.

The concept of majorization can be applied to pairs of matrices in $\mathcal{M}_d(\mathbb{C})$ by applying it to the vectors containing either the singular values or (in particular, for Hermitian matrices) the eigenvalues in decreasing order. Let for instance $s(A), s(B) \in \mathbb{R}^d$ be the vectors containing the singular values of two arbitrary complex matrices $A, B \in \mathcal{M}_d(\mathbb{C})$. By Ky Fan dominance (Thm.8.2) we get that

$$s(A) \succ s(B) \Leftrightarrow \|A\|_1 = \|B\|_1 \text{ and } \|A\| \geq \|B\| \text{ for all u.i. norms.} \quad (8.20)$$

In the following we will apply majorization to Hermitian matrices and write $A \succ B$ iff the corresponding majorization relation holds for the eigenvalues of A and B .

Theorem 8.7 (Majorization between density matrices) *Let ρ, σ be two density matrices in $\mathcal{M}_d(\mathbb{C})$. Then the following are equivalent:*

1. $\rho \succ \sigma$,
2. $\text{tr}[f(\rho)] \geq \text{tr}[f(\sigma)]$ for all convex functions $f : \mathbb{R} \rightarrow \mathbb{R}$,
3. there exists a set of unitaries $\{U_i \in \mathcal{M}_d(\mathbb{C})\}$ and probabilities p_i such that

$$\sigma = \sum_i p_i U_i \rho U_i^\dagger. \quad (8.21)$$

PROOF The equivalence 1. $\leftrightarrow$ 2. is an immediate application of Thm.8.4. For 3. $\rightarrow$ 1. we can w.l.o.g. assume that ρ and σ are both diagonal since the diagonalizing unitaries can be incorporated in the U_i 's. Then $\sigma_{kk} = \sum_l M_{kl} \rho_{ll}$ where $M_{kl} = \sum_i p_i |\langle k|U_i|l\rangle|^2$ is doubly stochastic and thus $\rho \succ \sigma$ by Thm.8.4. For the converse 1. $\rightarrow$ 3. we again use the freedom of assuming the ρ, σ are both diagonal. Then by Thm.8.4 and Birkhoff's Thm.8.6 we can write $\sigma_{kk} = \sum_{i,k,l} p_i \langle k|P_i|l\rangle^2 \rho_{ll}$ where the P_i 's are permutations and the p_i 's probabilities. Written in terms of the matrices this is $\sigma = \sum_i p_i P_i \rho P_i^T$ which completes the proof. $\square$

A similar result holds if we go from a fixed pair of density operators to all pairs which are related by a positive map:

Theorem 8.8 (Majorization and doubly stochastic positive maps) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a positive linear map. Then the following are equivalent:*

1. *T is trace preserving and unital,*
2. *For all density operators ρ it holds that $\rho \succ T(\rho)$,*
3. *For all density operators ρ and convex functions $f : \mathbb{R} \rightarrow \mathbb{R}$ it holds that $\text{tr}[f(\rho)] \geq \text{tr}[f(T(\rho))]$.*

PROOF The equivalence 2. $\leftrightarrow$ 3. follows from Thm.8.4 in the same way as it was used for the previous Thm.8.7. For 2. $\rightarrow$ 1. note that the trace preserving property is implied by the equality condition in the definition of majorization (see Eq.(8.16)). Unitality follows from assuming that $\mathbb{1} \succ T(\mathbb{1})$ together with the fact that the maximally mixed distribution only majorizes itself (see Eq.(8.17)). Finally, in order to show 1. $\rightarrow$ 2. let us consider an arbitrary density operator ρ with eigenvalue decomposition $\rho = V^\dagger \lambda(\rho) V$ and similarly $T(\rho) = U^\dagger \lambda(T(\rho)) U$, where $U, V \in \mathcal{M}_d(\mathbb{C})$ are unitaries and the λ 's denote the diagonal matrices containing the respective eigenvalues. Define a map $T' : \mathcal{M}_d \rightarrow \mathcal{M}_d$ via $T'(X) = UT(VXV^\dagger)U^\dagger$. By construction we have now $\lambda(T(\rho)) = T'(\lambda(\rho))$ which becomes $\lambda(T(\rho))_{ii} = \sum_j M_{ij} \lambda(\rho)_{jj}$ if we introduce $M_{ij} := \langle i | T'(|j\rangle\langle j|) | i \rangle$. The assertion now follows from the fact that M is a doubly stochastic matrix, which is in turn implied by T' being a trace-preserving, unital and positive map. $\square$

As a consequence we get that doubly stochastic quantum channels are exactly those for which the entropy is non-decreasing:

Corollary 8.1 (Entropy increasing maps) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a positive and trace-preserving linear map and $h : \mathbb{R} \rightarrow \mathbb{R}$ any strictly concave function.⁸ Then T is unital iff for all density operators $\rho \in \mathcal{M}_d(\mathbb{C})$ we have that*

$$\text{tr}[h(T(\rho))] \geq \text{tr}[h(\rho)]. \quad (8.22)$$

PROOF If T is unital, then the assertion follows from Thm.8.8 for every (not necessarily strictly) concave function h . Conversely, if h is strictly concave, then $\text{tr}[h(\rho)]$ has $\rho = \mathbb{1}/d$ as a unique maximizer within the set of density operators. So if we assume Eq.(8.22), then necessarily $T(\mathbb{1}) = \mathbb{1}$. $\square$

We will now use the developed tools in order to formalize the statement that any classical probability distribution obtained from measuring a quantum state is always at least as mixed as the density operator from which it has been obtained:

Theorem 8.9 (Measurements and convex functions)

Let $\{P_i \in \mathcal{M}_d(\mathbb{C})\}_{i=1,\dots,m}$ with $m \geq d$ be a POVM (i.e., $P_i \geq 0$ and $\sum_i P_i = \mathbb{1}$)

⁸‘Strictly concave’ means that for all $x_1 \neq x_2$ and $\lambda \in (0, 1)$ it holds that $h(\lambda x_1 + (1 - \lambda)x_2) > \lambda h(x_1) + (1 - \lambda)h(x_2)$.

and $\rho \in \mathcal{M}_d(\mathbb{C})$ a density operator. Assume that the POVM is symmetric in the sense that $\text{tr}[P_i] = d/m$ and denote the probabilities obtained from a measurement on ρ by $p_i := \text{tr}[P_i\rho]$. For every convex function $f : \mathbb{R} \rightarrow \mathbb{R}$ which satisfies $f(0) \leq 0$ we have

$$\text{tr}[f(\rho)] \geq \sum_{i=1}^m f(p_i). \quad (8.23)$$

For $m = d$ the same holds true without $f(0) \leq 0$ being required.

PROOF We want to make use of the implication 1. $\rightarrow$ 3. in Thm.8.8. To this end we first embed everything in an m -dimensional space and set for instance $\rho' := \rho \oplus 0 \in \mathcal{M}_m$. Moreover, we define a linear map $T : \mathcal{M}_m \rightarrow \mathcal{M}_m$ via

$$T(X) := \frac{\mathbb{1}_m}{m} \text{tr}[X(0 \oplus \mathbb{1}_{m-d})] + \sum_{i=1}^m |i\rangle\langle i| \text{tr}[X(P_i \oplus 0)]. \quad (8.24)$$

Note that T is positive, unital and trace-preserving as required by Thm.8.8. Then

$$\begin{aligned} \text{tr}[f(\rho)] &\geq \text{tr}[f(\rho')] \geq \text{tr}[f(T(\rho'))], \\ &= \sum_{i=1}^m f(p_i), \end{aligned} \quad (8.25)$$

where the first inequality comes from $f(0) \leq 0$ which is only required if $m > d$ since $\rho' = \rho$ for $m = d$. $\square$

Note that for von Neumann measurements, i.e., when the P_i 's are one-dimensional orthogonal projections, the above theorem essentially reduces to saying that the eigenvalues of ρ majorize the diagonal entries (see Eq.(8.18)). We also remark that Thm.8.9 is no longer true in general if we drop any of the three assumptions: $f(0) \leq 0$, $m \geq d$ or $\text{tr}[P_i] = d/m$.

8.4 Divergences and quasi-relative entropies

In classical probability theory, one way of quantifying how two probability distributions p and q differ from each other (other than using norms and metrics) is to use *Csiszar's f -divergence* measures

$$C_f(p|q) := \sum_{x \in X} q_x f\left(\frac{p_x}{q_x}\right), \quad (8.26)$$

where $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ is any continuous convex function. Every such C_f is jointly convex in its arguments and monotone in the sense that $C_f(p|q) \geq C_f(t(p)|t(q))$ for any map $t : p \mapsto p'$ characterized by a partitioning of the set of events X into disjoint X_i 's so that $p'_i = \sum_{x \in X_i} p_x$. If $f(1) = 0$ then $C_f(p|p) = 0$ and if $f(x)$ is strictly convex at $x = 1$ then $C_f(p|q) = 0$ implies $p = q$. Prominent examples of f -divergences are:

1. *Kullback Leibler divergence* or *relative entropy* for $f(x) = x \log x$:

$$Q_{KL}(p|q) = \sum_x p_x \log \frac{p_x}{q_x}, \quad (8.27)$$

2. *Relative α -entropy*⁹ for $f(x) = (1 - x^\alpha)/(\alpha(1 - \alpha))$ with $\alpha \in (0, 1)$:

$$Q_\alpha(p|q) = \left(1 - \sum_x p_x^\alpha q_x^{1-\alpha}\right)/(\alpha(1 - \alpha)). \quad (8.28)$$

This converges to the Kullback Leibler divergence as $\lim_{\alpha \rightarrow 0} Q_\alpha(p|q) = Q_{KL}(p|q)$ and $\lim_{\alpha \rightarrow 1} Q_\alpha(p|q) = Q_{KL}(p|q)$.

3. χ^2 -divergence for $f(x) = x^2 - 1$:

$$Q_{\chi^2}(p|q) = \sum_x \frac{(p_x - q_x)^2}{q_x} = \left(\sum_x \frac{p_x^2}{q_x}\right) - 1. \quad (8.29)$$

Note that by Taylor expansion the χ^2 -divergence locally approximates every f -divergence which is twice differentiable at $x = 1$. More precisely $C_f(p|q)/C_{\chi^2}(p|q) \rightarrow \frac{1}{2}f''(1)$ as $p \rightarrow q$.

4. *Hellinger divergence* for $f(x) = (\sqrt{x} - 1)^2/2$:

$$Q_{h^2}(p|q) = \left(1 - \sum_x \sqrt{p_x q_x}\right) = \frac{1}{2} \sum_x (\sqrt{p_x} - \sqrt{q_x})^2. \quad (8.30)$$

The square root of Q_{h^2} is a metric since it is proportional to the norm difference in l_2 .

5. l_1 -norm is the only bare metric among the f -divergences and is obtained for $f(x) = |x - 1|$:

$$Q_{l_1}(p|q) = \sum_x |p_x - q_x|. \quad (8.31)$$

In the following we will discuss generalizations of these ‘distance’ measures to density operators.

8.4.1 χ^2 divergence

A simple generalization of the χ^2 divergence in Eq.(8.29) to the non-commutative context of density operators ρ and σ is

$$\begin{aligned} \chi_\alpha^2(\rho|\sigma) &:= \operatorname{tr} [(\rho - \sigma)\sigma^{-\alpha}(\rho - \sigma)\sigma^{\alpha-1}], \quad \alpha \in \mathbb{R} \\ &= \operatorname{tr} [\rho\sigma^{-\alpha}\rho\sigma^{\alpha-1}] - 1. \end{aligned} \quad (8.32)$$

⁹In fact, there are many variants of relative α -entropies in the literature—most of them are slight modifications of others. The *relative Renyi entropy* for instance is usually defined as $S_\alpha(p|q) = (\alpha - 1)^{-1} \log \sum_x p_x^\alpha q_x^{1-\alpha}$ with $\alpha \geq 0$. In the limit $\alpha \rightarrow 1$ this becomes $Q_{KL}(p|q)$ again.

8.4.2 Quasi-relative entropies

The relative entropy has generalizing relatives in very much the same way as the von Neumann entropy has Renyi and Tsallis type cousins. The most prominent ones are functions depending on the one-parameter family $\text{tr} [\rho^\alpha \sigma^{1-\alpha}]$ where $\alpha \in \mathbb{R}$. These quantities appear naturally in the context of hypothesis testing (see Sec.8.5). The Tsallis relative entropy $S_{(\alpha)}$ and the Renyi relative entropy S_α are defined as

$$S_{(\alpha)}(\rho|\sigma) := \frac{1}{1-\alpha} (1 - \text{tr} [\rho^\alpha \sigma^{1-\alpha}]), \quad (8.33)$$

$$S_\alpha(\rho|\sigma) := \frac{1}{\alpha-1} \log \text{tr} [\rho^\alpha \sigma^{1-\alpha}]. \quad (8.34)$$

Both are non-negative for $\alpha \geq 0$ and coincide with the relative entropy $S(\rho|\sigma) = \text{tr} [\rho(\log \rho - \log \sigma)]$ in the limit $\alpha \rightarrow 1$.¹⁰ One of the main properties of these generalized relative entropies follows from Thm.5.15 which implies that the map $(\rho, \sigma) \mapsto \text{tr} [\rho^\alpha \sigma^{1-\alpha}]$ is jointly concave for $\alpha \in [0, 1]$ and jointly convex for $\alpha \in [-1, 0]$. Exploiting that joint convexity implies monotonicity under completely positive trace-preserving maps (see Thm.5.16) we arrive at:

Theorem 8.10 (Monotonicity of quasi-relative entropies) *Let $\rho, \sigma \in \mathcal{M}_d(\mathbb{C})$ be two density matrices. Then for all completely positive trace-preserving linear maps $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ and $\alpha \in [0, 1]$*

$$\text{tr} [\rho^\alpha \sigma^{1-\alpha}] \leq \text{tr} [T(\rho)^\alpha T(\sigma)^{1-\alpha}], \quad (8.35)$$

and the reverse inequality holds for $\alpha \in [-1, 0]$.

This means that for $\alpha \in [0, 1]$ both Renyi and Tsallis relative entropy are non-increasing under the action of a quantum channel.

A useful tool for deriving properties of the quasi-relative entropies in Eqs.(8.33,8.34) is a simple method which relates them to their classical counterparts: consider the spectral decompositions $\rho = \sum_i \lambda_i |x_i\rangle\langle x_i|$ and $\sigma = \sum_j \mu_j |y_j\rangle\langle y_j|$ and define $p_{ij} := \lambda_i |\langle x_i | y_j \rangle|^2$ and $q_{ij} := \mu_j |\langle x_i | y_j \rangle|^2$. Then $p, q \in \mathbb{R}^{2d}$ are classical probability distributions and straight forward calculation shows that

$$\text{tr} [\rho^\alpha \sigma^{1-\alpha}] = \sum_x q_x \left(\frac{p_x}{q_x} \right)^\alpha, \quad (8.36)$$

$$S(\rho|\sigma) = \sum_x p_x \log \frac{p_x}{q_x}. \quad (8.37)$$

That is, under the mapping $(\rho, \sigma) \rightarrow (p, q)$ the quantum relative entropies coincide with their classical analogues. This allows us to easily prove some of their properties—or simply to import them from the world of classical probability distributions:

¹⁰Outside the interval $\alpha \in [0, 1]$ inverses are taken on the support and the relative entropies are set to ∞ if the kernel of the operator to be inverted is not contained in the kernel of the other operator.

Theorem 8.11 (Relation among relative entropies) *Let $\rho, \sigma \in \mathcal{M}_d(\mathbb{C})$ be a pair of positive semi-definite operators and $S_{(\alpha)}(\rho|\sigma) := (\text{tr} [\rho - \rho^\alpha \sigma^{1-\alpha}]) / (1 - \alpha)$. Then for all $\alpha \leq 1$ and $\beta \geq 1$ we have*

$$S_{(\alpha)}(\rho|\sigma) \leq S(\rho|\sigma) \leq S_{(\beta)}(\rho|\sigma). \quad (8.38)$$

PROOF The result follows from the fact that for all positive numbers t and s it holds that $(1 - t^{-s})/s \leq \log t \leq (t^s - 1)/s$. This gives

$$\frac{p_x}{s} (1 - p_x^{-s} q_x^s) \leq p_x \log \frac{p_x}{q_x} \leq \frac{p_x}{s} (p_x^s q_x^{-s} - 1). \quad (8.39)$$

Using the relations in Eqs.(8.36,8.37) the desired result follows by setting $s = 1 - \alpha$ on the l.h.s. and $s = \beta - 1$ on the right. $\square$

Note that other properties and uses of the functional $\text{tr} [\rho^\alpha \sigma^{1-\alpha}]$ will be discussed in Sec.8.5 in the context of hypothesis testing.

8.4.3 Fidelity

8.5 Hypothesis testing

Hypothesis testing is the simplest task in the context of statistical inference: given two possibilities, find out which one is actually realized by performing a statistical test. General examples are questions of the form “was there a Higgs particle generated?” or “can we identify a tumor in our tomographic data?”. In our context the two hypotheses correspond to two density operators and we will see that several of the previously discussed distance measures for density operators will get an operational meaning as a means of quantifying how distinguishable the two states are.

Suppose we are given one out of two possible sources which produce states characterized by density operators ρ and σ respectively. Assume further that the a priori probability for ρ is p so that $(1 - p)$ is the prior for σ . Imagine we want to decide whether the actual source produces ρ or σ by performing a single two-outcome measurement with effect operators $\{P, \mathbb{1} - P\}$. We then assign one of the measurement outcomes, say the one corresponding to P , to the hypothesis ρ while the other outcome leads us to guess σ . In this way, the average probability for a false conclusion will be

$$e(P) := p \text{tr} [\rho(\mathbb{1} - P)] + (1 - p) \text{tr} [\sigma P]. \quad (8.40)$$

The following tells us how to minimize this average error probability and provides the optimal measurement:

Theorem 8.12 (Quantum Neyman-Pearson) *Within the above context the error probability satisfies the bound*

$$e(P) \geq \frac{1}{2} (1 - \|p\rho - (1 - p)\sigma\|_1), \quad (8.41)$$

where equality is achieved iff P is a projection onto the positive part of $[p\rho - (1-p)\sigma]$.

PROOF We have to minimize $e(P)$ w.r.t. all $0 \leq P \leq \mathbb{1}$, i.e.,

$$\min_P e(P) = p - \max_P \operatorname{tr} [(p\rho - (1-p)\sigma)P]. \quad (8.42)$$

Decomposing $(p\rho - (1-p)\sigma) =: A$ into orthogonal positive and negative parts as $A = A_+ - A_-$ it becomes clear that the maximum is attained iff $PA_+ = A_+$ and $PA_- = 0$. Eq.(8.41) then follows by using that for Hermitian operators $\operatorname{tr}[A_+] = (\|A\|_1 + \operatorname{tr}[A])/2$. $\square$

This gives a simple operational meaning to the trace norm distance between density operators: it quantifies how good the states can be distinguished by a single measurement.

Let us now go one step further and assume that we can measure not only on a single state but on $n \in \mathbb{N}$ copies of the system. That is, we let the source produce n identical state (either ρ or σ) before we perform a global measurement. According to the above Thm.8.12 an optimal measurement leads to an error probability

$$e_n := \frac{1}{n} \left(1 - \|p\rho^{\otimes n} - (1-p)\sigma^{\otimes n}\|_1 \right). \quad (8.43)$$

This turns out to decrease exponentially as $\exp(-\xi n)$ at a rate which is asymptotically independent of p (unless $p = 0, 1$) and given in the following theorem:

Theorem 8.13 (Quantum Chernoff bound) *For every non-trivial a priori probability $p \neq 0, 1$ an optimal sequence of measurements yields*

$$\xi := \lim_{n \rightarrow \infty} \left(-\frac{1}{n} \log e_n \right) = -\log \left(\inf_{s \in [0,1]} \operatorname{tr} [\rho^{1-s} \sigma^s] \right). \quad (8.44)$$

In some contexts it is natural to treat the two errors (1. we guess ρ while it was σ and 2. we guess σ while it was ρ) in an asymmetric way.¹¹ Assume we again perform a two-outcome measurement described by effect operators $P_n \in \mathcal{M}_{d^n}(\mathbb{C})$ and $(\mathbb{1} - P_n)$ on n copies of the system. Consider the task of minimizing the error probability

$$\beta_n(P_n) := \operatorname{tr} [\sigma^{\otimes n} P_n] \quad \text{under the constraint} \quad \operatorname{tr} [\rho^{\otimes n} (\mathbb{1} - P_n)] \leq \epsilon. \quad (8.45)$$

For a sequence of optimal measurements the error β_n will again decrease exponentially with n and the rate will turn out to be the relative entropy $S(\rho|\sigma)$. The analogous classical result is sometimes called *Stein's Lemma* where the Kullback-Leibler divergence appears as the optimal rate function. Before we come to the quantum version of this result we state a crucial ingredient for its proof which eventually enables us again to reduce the quantum to the classical case:

¹¹In classical statistical analysis this is natural for instance in many medical contexts: there we want to be sure not to overlook a certain disease, i.e., the probability for a false negative test should be small while one may accept a larger fraction of false positive tests.

Theorem 8.14 (Hiai-Petz) *Let $\rho, \sigma \in \mathcal{M}_d(\mathbb{C})$ be two density operators and $n \in \mathbb{N}$. Define a map $T : \mathcal{M}_{d^n}(\mathbb{C}) \rightarrow \mathcal{M}_{d^n}(\mathbb{C})$ via $T(X) := \sum_{i=1}^k P_i X P_i$ from the spectral decomposition $\sigma^{\otimes n} = \sum_{i=1}^k \lambda_i P_i$ where the sum runs over k distinct eigenvalues and the P_i 's project onto the corresponding eigenspaces. Then*

$$S(T(\rho^{\otimes n})|\sigma^{\otimes n}) \leq nS(\rho|\sigma) \leq S(T(\rho^{\otimes n})|\sigma^{\otimes n}) + d \log(n+1), \quad (8.46)$$

where $S(\rho|\sigma) = \text{tr}[\rho(\log \rho - \log \sigma)]$ is the relative entropy.

PROOF The left inequality follows from monotonicity of the relative entropy under trace-preserving completely positive maps together with additivity $S(\rho^{\otimes n}|\sigma^{\otimes n}) = nS(\rho|\sigma)$ and the invariance $T(\sigma^{\otimes n}) = \sigma^{\otimes n}$.

For the right inequality we use again additivity and proceed as follows:

$$n S(\rho|\sigma) \leq \text{tr}[\rho^{\otimes n}(\log T(\rho^{\otimes n}) - \log \sigma^{\otimes n})] + \log k \quad (8.47)$$

$$\leq S(T(\rho^{\otimes n})|\sigma^{\otimes n}) + d \log(n+1). \quad (8.48)$$

Here the first inequality uses $\log \rho^{\otimes n} \leq \log T(\rho^{\otimes n}) + \log k$ which follows from Lemma 8.2 together with the operator monotonicity of the logarithm. The second inequality exploits that (i) T is a conditional expectation (see Eq. (1.43)), so that in particular $\text{tr}[AT(B)] = \text{tr}[T(A)T(B)]$ for any A, B and (ii) the simple combinatorial bound $k \leq (n+1)^d$. $\square$

Note that Eq.(8.46) implies in particular that

$$S(\rho|\sigma) = \lim_{n \rightarrow \infty} \frac{1}{n} S(T(\rho^{\otimes n})|\sigma^{\otimes n}). \quad (8.49)$$

Moreover since $\sigma^{\otimes n}$ commutes with $T(\rho^{\otimes n})$ the relative entropy on the r.h.s. equals the one of the classical probability distributions which are obtained from measuring the two states in the basis in which they are simultaneously diagonal. Since the relative entropy (or Kullback-Leibler divergence) is the optimal rate function appearing in the classical Stein's lemma, the above Thm.8.14 implies that $S(\rho|\sigma)$ is an achievable rate in the quantum context as well. The following shows that it is indeed the optimal rate:

Theorem 8.15 (Quantum Stein Lemma) *Consider the task of distinguishing two quantum states $\rho, \sigma \in \mathcal{M}_d(\mathbb{C})$. Let β_n be the error probability as defined in Eq.(8.45), minimized over all measurements. For every $\epsilon \in (0, 1)$ we have that*

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log \beta_n = S(\rho|\sigma). \quad (8.50)$$

PROOF As discussed above achievability of the rate $S(\rho|\sigma)$ follows from Thm.8.14 together with the classical Stein's lemma. What remains to prove is thus an upper bound on the optimal rate, i.e., a lower bound on the error probability

β_n . To this end we apply Lemma 8.1 to $A = \rho^{\otimes n}$ and $B = e^{\lambda n} \sigma^{\otimes n}$ for some $\lambda \in \mathbb{R}$ to be chosen later. For $s \in [0, 1]$ this gives

$$e^{-s\lambda n} \text{tr} [\rho^{1+s} \sigma^{-s}]^n \geq \text{tr} [(\rho^{\otimes n} - e^{\lambda n} \sigma^{\otimes n}) P_n] \quad (8.51)$$

$$\geq (1 - \epsilon) - e^{n\lambda} \beta_n(P_n), \quad (8.52)$$

where the second inequality follows from Eq.(8.45). Rewriting these inequalities we obtain

$$\beta_n \geq e^{-n\lambda} \left[(1 - \epsilon) - e^{-n(\lambda s - f(s))} \right], \quad (8.53)$$

where we set $f(s) := \log \text{tr} [\rho^{1+s} \sigma^{-s}]$. Since $f(0) = 0$ and $f'(0) = S(\rho|\sigma)$ the choice $\lambda = S(\rho|\sigma) + \delta$ will for any $\delta > 0$ guarantee that there is an $s \in (0, 1]$ such that $\lambda s > f(s)$. Thus, $\lim_{n \rightarrow \infty} -\frac{1}{n} \beta_n \leq S(\rho|\sigma) + \delta$ and since this holds for arbitrary $\delta > 0$ the relative entropy is indeed the optimal asymptotic rate. $\square$

Lemma 8.1 *Let $A, B \in \mathcal{M}_d(\mathbb{C})$ be two positive semi-definite operators and denote by $(A - B)_+$ the positive part of the Hermitian matrix $(A - B)$. Then for all $s \in [0, 1]$:*

$$\|A - B\|_1 \geq \text{tr} [A + B] - 2\text{tr} [A^s B^{1-s}], \quad \text{and} \quad (8.54)$$

$$\text{tr} [(A - B)_+] \leq \text{tr} [A^{1+s} B^{-s}]. \quad (8.55)$$

PROOF In order to prove Eq.(8.55) we show first that it is sufficient to consider diagonal A and B . To this end we exploit that $\text{tr} [A^{1+s} B^{-s}]$ is non-increasing under completely positive, trace-preserving maps (see e.g. Thm.8.10). We apply this to the map $T(\cdot) := \sum_i P_i \cdot P_i$ which is constructed from the spectral decomposition $(A - B) = \sum_i \lambda_i P_i$ where the P_i 's are one-dimensional projections. Then since $\text{tr} [(A - B)_+] = \text{tr} [(T(A) - T(B))_+]$ and now $[T(A), T(B)] = 0$ it is indeed sufficient to consider diagonal A and B . For those the assertion follows from the simple inequality $a - b \leq a(a/b)^s$ which holds for numbers $a \geq b \geq 0$. $\square$

Lemma 8.2 *Let $\rho \in \mathcal{M}_d(\mathbb{C})$ be a density operator and $\{P_i \in \mathcal{M}_d(\mathbb{C})\}_{i=1, \dots, k}$ a POVM where each P_i is a projection. If $T(X) := \sum_{i=1}^k P_i X P_i$, then*

$$\rho \leq kT(\rho). \quad (8.56)$$

PROOF Due to linearity it is sufficient to consider pure states $\rho = |\psi\rangle\langle\psi|$. Then for every $|\phi\rangle \in \mathbb{C}^d$

$$\langle\phi|kT(\rho) - \rho|\phi\rangle = \left(k \sum_{i=1}^k |\langle\phi|P_i|\psi\rangle|^2 \right) - |\langle\phi|\sum_{i=1}^k P_i|\psi\rangle|^2 \geq 0, \quad (8.57)$$

where the inequality follows from Chauchy-Schwarz when applied to a vector with components $\langle\phi|P_i|\psi\rangle$ and the other with components k -times 1. $\square$

8.6 Hilbert's projective metric

Let $\mathcal{V}$ be a real, finite-dimensional vector space (e.g., the space of Hermitian matrices in $\mathcal{M}_d(\mathbb{C})$). A *convex cone* $\mathcal{C} \subset \mathcal{V}$ is a subset for which $\alpha\mathcal{C} + \beta\mathcal{C} \subseteq \mathcal{C}$ for all $\alpha, \beta \geq 0$. We will call a convex cone in $\mathcal{V}$ a *proper cone* if it is closed, has a non-empty interior and satisfies $\mathcal{C} \cap (-\mathcal{C}) = \{0\}$. Examples of proper cones are sets of non-negative polynomials, vectors or matrices with non-negative entries, positive semi-definite matrices, and the cones of un-normalized density matrices which are separable, have a positive partial transpose, are undistillable, etc.

Recall that the *dual cone* is defined as $\mathcal{C}^* := \{c \in \mathcal{V} | \forall v \in \mathcal{C} : \langle v | c \rangle \geq 0\}$ and that by the *bipolar theorem* $\mathcal{C}^{**} = \mathcal{C}$ for closed convex cones. This implies that

$$v \in \mathcal{C} \Leftrightarrow \langle v | c \rangle \geq 0 \text{ for all } c \in \mathcal{C}^*. \quad (8.58)$$

Every proper cone induces a (partial) order relation in $\mathcal{V}$ and we will write $a \succeq_{\mathcal{C}} b$ meaning $a - b \in \mathcal{C}$. We will omit the subscript $\mathcal{C}$ if the cone $\mathcal{C}$ under consideration is the set of positive semi-definite matrices. Also note that in this case we have $\mathcal{C} = \mathcal{C}^*$.

For every pair of non-zero elements $a, b \in \mathcal{C}$ define

$$\sup(a/b) := \sup_{c \in \mathcal{C}^*} \frac{\langle a | c \rangle}{\langle b | c \rangle}, \quad \inf(a/b) := \inf_{c \in \mathcal{C}^*} \frac{\langle a | c \rangle}{\langle b | c \rangle}. \quad (8.59)$$

By construction $\sup(a/b) \geq \inf(a/b) \geq 0$ and $\sup(a/b) = 1/\inf(b/a)$. If $\mathcal{C}$ (and consequently also $\mathcal{C}^*$) is a proper cone we can use Eq.(8.58) and rewrite

$$\sup(a/b) = \inf\{\lambda \in \mathbb{R} | a \leq_{\mathcal{C}} \lambda b\}, \quad (8.60)$$

$$\inf(a/b) = \sup\{\lambda \in \mathbb{R} | \lambda b \leq_{\mathcal{C}} a\}. \quad (8.61)$$

This implies that $\inf(a/b)b \leq_{\mathcal{C}} a \leq_{\mathcal{C}} \sup(a/b)b$, where the latter makes sense only if $\sup(a/b)$ is finite. In other words, Eq.(8.59) provides the factors by which b has to be rescaled at least in order to become larger or smaller than a .

Hilbert's projective metric is then defined for non-zero $a, b \in \mathcal{C}$ as

$$D(a, b) := \ln [\sup(a/b) \sup(b/a)]. \quad (8.62)$$

Obviously, D is symmetric, non-negative and satisfies $D(a, \beta b) = D(a, b)$ for all $\beta > 0$. That is, D depends only on the 'direction' of its arguments. Since it turns out to fulfill the triangle inequality and $D(a, b) = 0$ implies that $a = \beta b$ for some $\beta > 0$, D is a *projective metric*. Hence, if we restrict the arguments a, b further to a subset which excludes multiples of elements (such as the unit sphere of a norm), then D becomes a metric on that space. With $\mathcal{C}$ the cone of positive semi-definite matrices D , for instance, is a metric on the set of density matrices. The following relates Hilbert's projective metric to metrics induced by norms.

Proposition 8.2 (Norm bound) *Consider a proper cone $\mathcal{C}$ in a real linear space. Let $A, B \in \mathcal{C}$ be non-zero elements and define $\nu := \inf(B/A)$ and $\mu := \inf(A/B)$. Then*

$$\begin{aligned} \|A - B\| &\leq |1 - \nu| \|A\| + |1 - \mu| \|B\| \\ &\quad + \min[|1 - \nu| \|A\|, |1 - \mu| \|B\|], \end{aligned} \quad (8.63)$$

for every norm for which $X \geq_{\mathcal{C}} Y \geq_{\mathcal{C}} 0$ implies $\|X\| \geq \|Y\|$.

PROOF By definition $B \geq_{\mathcal{C}} \nu A$ which implies $A - B \leq_{\mathcal{C}} (1 - \nu)A$ and similar holds for μ if only A and B are interchanged. This guarantees the existence of positive elements $P_\nu, P_\mu \geq_{\mathcal{C}} 0$ such that

$$A - B + P_\nu = (1 - \nu)A \quad \text{and} \quad B - A + P_\mu = (1 - \mu)B. \quad (8.64)$$

Using the triangular inequality for the norm $\|A - B\| \leq \|A - B + P_\nu\| + \|P_\nu\|$ and exploiting its assumed monotonicity by bounding $\|P_\nu\| \leq \|P_\nu + P_\mu\|$, Eq.(8.64) then leads to

$$\|A - B\| \leq \|A - B + P_\nu\| + \|P_\nu + P_\mu\| \quad (8.65)$$

$$\leq 2|1 - \nu| \|A\| + |1 - \mu| \|B\|. \quad (8.66)$$

Since the same reasoning and thus the same inequality applies with $\mu \leftrightarrow \nu$ and A and B interchanged, Eq.(8.63) then follows from taking the smaller bound. $\square$

Corollary 8.2 (Norms vs. Hilbert's projective metric) *With the assumptions of Prop.8.2 and the additional requirement that $\mu, \nu \leq 1$ we have*

$$\|A - B\| \leq 3 \left(\frac{\|A\| + \|B\|}{2} - \sqrt{\|A\| \|B\|} e^{-D(A,B)/2} \right). \quad (8.67)$$

PROOF Instead of taking the minimum in Eq.(8.63) we take the average and then use that for positive numbers $(a + b)/2 \geq \sqrt{ab}$. In this way we obtain

$$\|A - B\| \leq \frac{3}{2} (\|A\| + \|B\| - (\nu\|A\| + \mu\|B\|)) \quad (8.68)$$

$$\leq 3 \left(\frac{\|A\| + \|B\|}{2} - \sqrt{\nu\|A\|\mu\|B\|} \right), \quad (8.69)$$

which leads the sought result by using $\inf(A/B) = 1/\sup(B/A)$ together with the definition Eq.(8.62). $\square$

Better bounds can be obtained by exploiting some additional structure. So let us now explicitly assume that $\mathcal{C}$ is the cone of positive semi-definite operators on a finite-dimensional Hilbert space and consider the trace norm:

Proposition 8.3 (Trace norm vs. Hilbert's projective metric) *Let $\rho_1, \rho_2 \in \mathcal{M}_d(\mathbb{C})$ be two density matrices. Then*

$$\frac{1}{2} \|\rho_1 - \rho_2\|_1 \leq [1 + \inf(\rho_2/\rho_1)]^{-1} - [1 + \sup(\rho_2/\rho_1)]^{-1} \quad (8.70)$$

$$\leq \tanh[D(\rho_1, \rho_2)/4]. \quad (8.71)$$

PROOF Recall that $\|\rho_1 - \rho_2\|_1/2 = \text{tr}[P(\rho_1 - \rho_2)]$ for some Hermitian projection P . Exploiting that $x(2-x) \leq 1$ for all $x \in [0, 2]$ and applying it to $x = \text{tr}[P(\rho_1 + \rho_2)]$ we obtain

$$\frac{1}{2}\|\rho_1 - \rho_2\|_1 \leq \frac{\text{tr}[P(\rho_1 - \rho_2)]}{\text{tr}[P(\rho_1 + \rho_2)](2 - \text{tr}[P(\rho_1 + \rho_2)])} \quad (8.72)$$

$$= (1 + \text{tr}[P\rho_2]/\text{tr}[P\rho_1])^{-1} - (1 + \text{tr}[P'\rho_2]/\text{tr}[P'\rho_1])^{-1} \quad (8.73)$$

where $P' = \mathbb{1} - P$ and the last equation follows from elementary algebra. Eq.(8.70) now follows from Eq.(8.73) by taking the supremum over all $P, P' \geq 0$ and inserting the definitions from Eq.(8.59).

In order to arrive at Eq.(8.71) we use that $\inf(\rho_2/\rho_1) = 1/\sup(\rho_1/\rho_2)$ and abbreviate $x := \sup(\rho_1/\rho_2)$, $y := \sup(\rho_2/\rho_1)$. In this way we can rewrite and bound the r.h.s. of Eq.(8.70) as

$$\frac{xy - 1}{(x+1)(y+1)} \leq \frac{\sqrt{xy} - 1}{\sqrt{xy} + 1}, \quad (8.74)$$

where the inequality follows from applying the inequality $(x+1)(y+1) \geq (\sqrt{xy} + 1)^2$, which holds for all $x, y \geq 0$. Using that according to definition (8.62) we have $xy = \exp[D(\rho_1, \rho_2)]$ then completes the proof. $\square$

Exploiting that $2\tanh(x/2) \leq x$ for $x \geq 0$ we obtain in particular

$$\|\rho_1 - \rho_2\|_1 \leq D(\rho_1, \rho_2)/2. \quad (8.75)$$

Note that a non-trivial lower bound on norms cannot exist: suppose that for some vector ψ we have $\langle \psi | \rho_1 | \psi \rangle > 0$ but $\langle \psi | \rho_2 | \psi \rangle = 0$. Then $\sup(\rho_1/\rho_2) = \infty$ according to Eq.(8.60) so that $D(\rho_1, \rho_2) = \infty$ although the two states could be arbitrarily close in norm. However, if the supports of ρ_1 and ρ_2 are the same, Hilbert's metric is finite and can be explicitly expressed in terms of operator norms:

Proposition 8.4 *Let $\rho_1, \rho_2 \in \mathcal{M}_d(\mathbb{C})$ be density matrices and consider the positive semi-definite cone $\mathcal{C}$. Then with $^{-1}$ denoting the pseudo inverse (inverse on the support) we have*

$$\sup(\rho_1/\rho_2) = \begin{cases} \|\rho_2^{-1/2}\rho_1\rho_2^{-1/2}\|_\infty, & \text{if } \text{supp}[\rho_1] \subseteq \text{supp}[\rho_2] \\ \infty & \text{otherwise} \end{cases} \quad (8.76)$$

$$\inf(\rho_1/\rho_2) = \begin{cases} \|\rho_1^{-1/2}\rho_2\rho_1^{-1/2}\|_\infty^{-1}, & \text{if } \text{supp}[\rho_2] \subseteq \text{supp}[\rho_1] \\ 0 & \text{otherwise} \end{cases} \quad (8.77)$$

$$D(\rho_1, \rho_2) = \begin{cases} \ln \left[\|\rho_1^{-1/2}\rho_2\rho_1^{-1/2}\|_\infty \|\rho_2^{-1/2}\rho_1\rho_2^{-1/2}\|_\infty \right], & \text{if } \text{supp}[\rho_2] = \text{supp}[\rho_1] \\ \infty & \text{otherwise} \end{cases}$$

PROOF We only have to prove the relation for $\sup(\rho_1/\rho_2)$ since this implies the other two by $\inf(\rho_1/\rho_2) = 1/\sup(\rho_2/\rho_1)$ and definition (8.62) respectively.

Assume that $\text{supp}[\rho_1] \not\subseteq \text{supp}[\rho_2]$. Then there is a vector ψ for which $\langle \psi | \rho_2 | \psi \rangle = 0$ while $\langle \psi | \rho_1 | \psi \rangle > 0$ so that the infimum in Eq.(8.60) is over an empty set and thus by the usual convention ∞ . If, however, $\text{supp}[\rho_1] \subseteq \text{supp}[\rho_2]$, then $\rho_1 \leq \lambda \rho_2$ is equivalent to $\rho_2^{-1/2} \rho_1 \rho_2^{-1/2} \leq \lambda \mathbb{1}$ and the smallest λ for which this holds is the operator norm. $\square$

An immediate consequence of this is a simple bound on the quantum χ^2 -divergence:

Corollary 8.3 *Let $\rho_1, \rho_2 \in \mathcal{M}_d(\mathbb{C})$ be density matrices. Then*

$$\chi_{1/2}^2(\rho_1 | \rho_2) \leq \sup(\rho_1 / \rho_2) - 1. \quad (8.78)$$

PROOF Recall the definition of the quantum χ^2 -divergence from Eq.(8.32): $\chi^2(\rho_1 | \rho_2) = \text{tr} [\rho_1 \rho_2^{-1/2} \rho_1 \rho_2^{-1/2}] - 1$. From here the assertion follows via Eq.(8.76) by using that $\text{tr} [\rho_1 \rho_2^{-1/2} \rho_1 \rho_2^{-1/2}] \leq \|\rho_2^{-1/2} \rho_1 \rho_2^{-1/2}\|_\infty$. $\square$

8.7 Contractivity and the increase of entropy

Trace norm

Theorem 8.16 (Trace norm contractivity) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ be a trace preserving positive linear map. Then for all Hermitian $H \in \mathcal{M}_d(\mathbb{C})$:*

$$\|T(H)\|_1 \leq \|H\|_1. \quad (8.79)$$

PROOF Let $T(H) = Q_+ - Q_-$ and $H = P_+ - P_-$ be decompositions into orthogonal parts $Q_\pm \geq 0$ and $P_\pm \geq 0$. Projecting the equation $Q_+ - Q_- = T(P_+) - T(P_-)$ onto the support space of Q_+ (or Q_-) and exploiting positivity of the involved expressions leads to $\text{tr}[Q_+] \leq \text{tr}[T(P_+)]$ (or $\text{tr}[Q_-] \leq \text{tr}[T(P_-)]$). Using the trace-preserving property of T then leads to $\text{tr}[Q_+ + Q_-] \leq \text{tr}[P_+ + P_-]$ which is a reformulation of Eq.(8.79). $\square$

Note that this means in particular that for all pairs of density operators ρ_1 and ρ_2 we have

$$\|T(\rho_1) - T(\rho_2)\|_1 \leq \|\rho_1 - \rho_2\|_1. \quad (8.80)$$

For a generic map T this inequality can in principle be sharpened in the sense that the r.h.s. is multiplied by a factor smaller than one. The best possible such factor which is independent of the ρ_i 's is called *contraction coefficient*. For the trace norm its computation is simplified due to the following:

Lemma 8.3 (Trace norm contraction coefficient) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ be a linear map. Then*

$$\sup_{\rho_1, \rho_2} \frac{\|T(\rho_1) - T(\rho_2)\|_1}{\|\rho_1 - \rho_2\|_1} = \frac{1}{2} \sup_{\psi \perp \phi} \|T(|\psi\rangle\langle\psi| - |\phi\rangle\langle\phi|)\|_1, \quad (8.81)$$

where the supremum on the l.h.s. is taken over all density matrices in $\mathcal{M}_d(\mathbb{C})$ and the one on the r.h.s. over all pairs of orthogonal state vectors in $\mathbb{C}^d$.

PROOF The r.h.s. in Eq.(8.81) is certainly a lower bound since the set over which the supremum is taken is reduced. So we have to show that it is also an upper bound. To this end, consider any difference of density operators with $\rho_1 - \rho_2 = P_+ - P_-$ being a decomposition into orthogonal positive and negative parts (i.e., $P_{\pm} \geq 0$ and $P_+P_- = 0$). By rescaling both numerator and denominator of $\|T(P_+ - P_-)\|_1 / \|P_+ - P_-\|_1$ we can achieve that $\text{tr}[P_{\pm}] = 1$ so that on the l.h.s. of Eq.(8.81) we can w.l.o.g. assume that the two density matrices have orthogonal supports. In this case $\|P_+ - P_-\|_1 = 2$. A convex decomposition then leads to $P_+ - P_- = \sum_i \lambda_i (|\psi_i\rangle\langle\psi_i| - |\phi_i\rangle\langle\phi_i|)$ with $\psi_i \perp \phi_i$ for all i . Exploiting convexity of the norm then finally leads to the equality in Eq.(8.81). $\square$

We will see below that non-trivial upper bounds for the trace-norm contraction coefficient, and thus improvements on inequality (8.80), can for instance be obtained from the *projective diameter* and the fidelity contraction coefficient. Another possibility to improve on Eq.(8.80) is to formalize the intuition that if T is strictly inside the cone of positive maps, then it should be a strict contraction. In the case of classical channels (i.e., stochastic matrices) this is the content of *Doebelin's theorem* of which a quantum counterpart looks as follows:

Theorem 8.17 (Quantum version of Doebelin's theorem) *Let $T, T' : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ be two trace-preserving and Hermiticity-preserving linear maps of which T' is such that $T'(X) = \text{tr}[X]Y$ for some $Y \in \mathcal{M}_{d'}(\mathbb{C})$. If $T - \epsilon T'$ is positive for some $\epsilon \geq 0$, then for all density matrices $\rho_1, \rho_2 \in \mathcal{M}_d(\mathbb{C})$:*

$$\|T(\rho_1) - T(\rho_2)\|_1 \leq (1 - \epsilon)\|\rho_1 - \rho_2\|_1. \quad (8.82)$$

PROOF We will use a variant of Eq.(8.10), namely that for any traceless Hermitian matrix X one can express the trace-norm as

$$\|X\|_1 = \sup_{0 \leq P \leq \mathbb{1}} 2\text{tr}[PX]. \quad (8.83)$$

Applying this to $X = T(\rho_1 - \rho_2)$ and using that $T'(\rho_1 - \rho_2) = 0$ we obtain

$$\|T(\rho_1) - T(\rho_2)\|_1 = \sup_{0 \leq P \leq \mathbb{1}} 2\text{tr}[(\rho_1 - \rho_2)(T - \epsilon T')^*(P)]. \quad (8.84)$$

Note that $P' := (T - \epsilon T')^*(P)$ is positive if, as assumed, $T - \epsilon T'$ is a positive map. Moreover, the fact that $P \leq \mathbb{1}$ together with the trace-preserving property of T and T' implies that $P' \leq (T - \epsilon T')^*(\mathbb{1}) = (1 - \epsilon)\mathbb{1}$ so that finally

$$\|T(\rho_1) - T(\rho_2)\|_1 \leq (1 - \epsilon) \sup_{0 \leq P \leq \mathbb{1}} 2\text{tr}[(\rho_1 - \rho_2)P], \quad (8.85)$$

from which the claimed inequality follows by exploiting Eq.(8.83) once again. $\square$

While positivity is difficult to decide, complete positivity is easily seen from the Choi-Jamolkowski operator (see Prop.2.1). So if $\tau = (T \otimes \text{id}_d)(|\Omega\rangle\langle\Omega|)$ is the Choi-Jamolkowski operator of T , then Eq.(8.82) holds for every $\epsilon \geq 0$ for which there is a Hermitian Y with $\text{tr}[Y] = 1$ so that

$$\tau \geq \frac{\epsilon}{d}(Y \otimes \mathbb{1}). \quad (8.86)$$

If T is highly contractive in the sense that it maps everything close to a highly mixed density operator ρ_∞ , then $Y = \rho_\infty$ can be a reasonable choice. Also note that for a given Y the largest ϵ satisfying Eq.(8.86) can be computed explicitly following the approach of Prop.8.4. Choosing $Y = \mathbb{1}/d$ enables us to relate the spectra of a channel and its Jamiolkowski state:¹²

Corollary 8.4 (Channel spectrum vs. Jamiolkowski spectrum) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a completely positive and trace-preserving linear map with corresponding Jamiolkowski state τ . If we denote by μ the largest modulus of eigenvalues of T which are strictly inside the unit disc and write $\lambda_{\min}(\tau)$ for the smallest eigenvalue of τ , then*

$$\mu \leq 1 - d^2 \lambda_{\min}(\tau). \quad (8.87)$$

PROOF Applying Thm.8.17 with $Y = \mathbb{1}/d$ to T^n for $n \in \mathbb{N}$ leads to $\|T^n(\rho_1 - \rho_2)\|_1 \leq (1 - d^2 \lambda_{\min}(\tau))^n \|\rho_1 - \rho_2\|_1$. As we will see in Prop.8.6 and Thm.8.23 below, $\sup_{\rho_1, \rho_2} \|T^n(\rho_1 - \rho_2)\|_1$ is, up to a constant, asymptotically lower bounded by μ^n so that Eq.(8.87) follows when considering $n \rightarrow \infty$. $\square$

Hilbert's projective metric While investigating Perron-Frobenius theory of positive maps [true?], Birkhoff [?] observed that all strictly positive maps are contractive w.r.t. Hilbert's projective metric. This allowed him in essence to view convergence under subsequent application of a positive map as a special case of the Banach contraction mapping theorem. We will use the notation and terminology of Sec.8.6 and assume throughout that all cones are embedded in finite-dimensional real vector spaces. In order to state Birkhoff's observation we need to define the *projective diameter* $\Delta(T)$ of a positive, i.e., cone-preserving, map $T : \mathcal{C} \rightarrow \mathcal{C}$ on a proper cone $\mathcal{C}$ (e.g., the cone of positive semi-definite matrices in $\mathcal{M}_d(\mathbb{C})$):

$$\Delta(T) := \sup_{a, b \in \mathcal{C}^\circ} D(T(a), T(b)), \quad (8.88)$$

where the supremum runs over all elements a, b in the interior of $\mathcal{C}$, denoted by $\mathcal{C}^\circ$.

Theorem 8.18 (Birkhoff's contraction theorem) *Let $\mathcal{C}$ be a proper cone and $T : \mathcal{C} \rightarrow \mathcal{C}$ a positive (i.e., cone-preserving) linear map. Then*

$$\sup_{a, b \in \mathcal{C}^\circ} \frac{D(T(a), T(b))}{D(a, b)} \leq \tanh(\Delta(T)/4). \quad (8.89)$$

In fact, Hilbert's metric is the essentially unique *projective metric* which allows to identify positive maps with contractive maps [?]. In order to formalize this, recall that a symmetric functional $D' : \mathcal{C} \times \mathcal{C} \rightarrow \mathbb{R}_+ \cup \infty$ is called *projective metric* if it satisfies $D'(a, b) \leq D'(a, c) + D'(c, b)$ for all $a, b, c \in \mathcal{C}$ and $D'(a, b) = 0$ iff $a = \lambda b$ for some $\lambda > 0$.

¹²For another relation of that kind see Prop.6.5.

Theorem 8.19 (Uniqueness of Hilbert's projective metric) *Let $\mathcal{C}$ be a proper cone and D' a projective metric such that every strictly positive linear map $T : \mathcal{C} \rightarrow \mathcal{C}^\circ$ is a strict contraction w.r.t. to D' . Then there exists a continuous and strictly increasing function $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $D' = f(D)$ where D is Hilbert's projective metric. Moreover, for any linear $T : \mathcal{C} \rightarrow \mathcal{C}^\circ$ we have*

$$\sup_{a,b \in \mathcal{C}^\circ} \frac{D(T(a), T(b))}{D(a, b)} \leq \sup_{a,b \in \mathcal{C}^\circ} \frac{D'(T(a), T(b))}{D'(a, b)}. \quad (8.90)$$

The projective diameter which appears in Birkhoff's theorem can in turn be lower bounded by the “second largest eigenvalue” [?]:

Theorem 8.20 (Spectral bound on projective diameter) *Let $\mathcal{C}$ be a proper cone and $T : \mathcal{C} \rightarrow \mathcal{C}$ a positive linear map for which $T(c) = c$ for some non-zero $c \in \mathcal{C}$. If $T(a) = \lambda a$ for some $\lambda \in \mathbb{C}$ and a not proportional to c , then*

$$|\lambda| \leq \tanh(\Delta(T)/4). \quad (8.91)$$

[I dropped the $\Delta < \infty$ and the $c \in \mathcal{C}^\circ$ requirement. Both seem superfluous .. proof? idea: perturb T in order to make it primitive and use that the spectrum changes continuously and that $\Delta(T) < \infty$ iff T is primitive (which should mean something like $T : \mathcal{C} \setminus \{0\} \rightarrow \mathcal{C}^\circ$)]

For the particular case of $\mathcal{C}$ being the cone of positive semi-definite matrices in $\mathcal{M}_d(\mathbb{C})$ we can show that the projective diameter also bounds the contraction w.r.t. the trace norm:

Theorem 8.21 (Trace norm contraction incl. projective diameter) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ be a positive, trace-preserving linear map. Then for all pairs of density matrices $\rho_1, \rho_2 \in \mathcal{M}_d(\mathbb{C})$*

$$\|T(\rho_1) - T(\rho_2)\|_1 \leq \|\rho_1 - \rho_2\|_1 \tanh(\Delta(T)/4). \quad (8.92)$$

PROOF Replacing $\rho_i \rightarrow T(\rho_i)$ in Prop.8.3 and taking the supremum over all full-rank density operators ρ_1, ρ_2 we obtain $\sup \|T(\rho_1 - \rho_2)\|/2 \leq \tanh(\Delta(T)/4)$. On the l.h.s. of this inequality we may as well take the supremum over all density matrices and then obtain a lower bound by restricting to orthogonal pure states. Following Lem.8.3 this lower bound equals the trace-norm contraction coefficient which then implies Eq.(8.92). $\square$

Birkhoff's theorem implies that if for two pairs of positive operators ρ_1, ρ_2 and ρ'_1, ρ'_2 there is a positive map $T : \rho_i \mapsto \rho'_i$, then $D(\rho_1, \rho_2) \geq D(\rho'_1, \rho'_2)$. In the following we will see that the converse is true as well: if D decreases, then there exists a completely positive map which does the mapping up to normalization. The main ingredient will be the following relation with the *condition number* which is defined for symmetric matrices as $\kappa(X) := \|X\|_\infty \|X^{-1}\|_\infty$.

Lemma 8.4 *Let $P \in \mathcal{M}_d(\mathbb{C})$ and $P' \in \mathcal{M}_{d'}(\mathbb{C})$ be positive definite operators. Then there exists a completely positive and unital linear map $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ such that $T(P) = \lambda P'$ for some $\lambda > 0$ iff*

$$\kappa(P) \geq \kappa(P'). \quad (8.93)$$

PROOF Let us first prove the “if part” and let all norms be operator norms. Note that $\kappa(P) = 1$ is equivalent to $P \propto \mathbb{1}$ for which the statement is trivially true (since $P' \propto \mathbb{1}$ as well). So let us assume that P is not proportional to the identity. Define two functions on spaces of symmetric matrices as $E_1(X) := \mathbb{1} - X/\|P\|$ and $E_2(X) := X - \mathbb{1}/\|P^{-1}\|$. By construction $E_i(P) \geq 0$ and $\text{span}\{E_1(P), E_2(P)\} = \text{span}\{P, \mathbb{1}\}$. Moreover, Eq.(8.93) implies that there is a $\lambda > 0$ such that

$$\frac{\|P\|}{\|P'\|} \geq \lambda \geq \frac{\|P'^{-1}\|}{\|P^{-1}\|}, \quad (8.94)$$

from which $E_i(\lambda P') \geq 0$ follows. Let ψ_1 (ψ_2) be an eigenvector corresponding to the smallest (largest) eigenvalue of P and define a map $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ via

$$T(X) = \sum_i \langle \psi_i | X | \psi_i \rangle \frac{E_i(\lambda P')}{\langle \psi_i | E_i(P) | \psi_i \rangle}. \quad (8.95)$$

Since $E_i(P)|\psi_j\rangle = 0$ for $i \neq j$ we have that $T : E_i(P) \mapsto E_i(\lambda P')$ and therefore $T(\mathbb{1}) = \mathbb{1}$ and $T(P) = \lambda P'$. Moreover, due to positivity of the involved expressions T is a completely positive map, as desired.

For the “only if part” assume that there is a positive unital map T which acts as $T(P) = \lambda P'$. Then, since $E_i(P) \geq 0$, we have to have that $T(E_i(P)) = E_i(\lambda P') \geq 0$. These two inequalities (for $i = 1, 2$) are, however, equivalent to Eq.(8.94) from which Eq.(8.93) follows. $\square$

As a consequence we obtain the claimed converse to Birkhoff’s theorem:

Theorem 8.22 *Let $\rho_1, \rho_2 \in \mathcal{M}_d(\mathbb{C})$ and $\rho'_1, \rho'_2 \in \mathcal{M}_{d'}(\mathbb{C})$ be two pairs of density matrices so that $\text{supp}(\rho_1) = \text{supp}(\rho_2)$. There exists a completely positive map $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ which acts as $T(\rho_i) = \lambda_i \rho'_i$ for some $\lambda_i > 0$ iff*

$$D(\rho_1, \rho_2) \geq D(\rho'_1, \rho'_2). \quad (8.96)$$

PROOF The ‘only if’ part follows from Birkhoff’s theorem. For the converse assume that Eq.(8.96) holds. The assumed equal support of ρ_1 and ρ_2 together with Eq.(8.96) implies that ρ'_1 and ρ'_2 have equal support as well. We can therefore restrict the input and output spaces to the respective support spaces and assume w.l.o.g. that all involved states have full rank. For any map T we can then define a map $\tilde{T}$ as

$$\tilde{T}(X) := \rho_1'^{-1/2} T(\rho_1^{1/2} X \rho_1^{1/2}) \rho_1'^{-1/2} / \lambda_1.$$

Note that T is completely positive iff $\tilde{T}$ is. Moreover, if we define $P := \rho_1^{-1/2} \rho_2 \rho_1^{-1/2}$, $P' := \rho_1'^{-1/2} \rho_2' \rho_1'^{-1/2}$ and $\lambda := \lambda_2 / \lambda_1$ then $T(\rho_i) = \lambda_i \rho'_i$ becomes equivalent to $\tilde{T}(P) = \lambda P'$ and $\tilde{T}(\mathbb{1}) = \mathbb{1}$. Hence, we can apply Lem.8.4 and by using Prop.8.4 we see that Eq.(8.96) is a reformulation of Eq.(8.93). $\square$

To conclude this discussion we will give an operational interpretation of this result: assume that for a finite set of pairs of density matrices $\{(\rho_i, \rho'_i)\}$ there is a completely positive map $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C})$ such that $T(\rho_i) =$

$\lambda_i \rho'_i$ for some $\lambda_i > 0$. Then we can construct a new map $\tilde{T} : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_{d'}(\mathbb{C}) \otimes \mathbb{Z}_2$ which is completely positive and trace-preserving and such that (i) $\rho_i \mapsto \rho'_i$ conditioned on outcome '1' on the ancillary two-level system, and (ii) this outcome is obtained with non-zero probability. More explicitly, this is obtained by

$$\tilde{T}(\rho) := cT(\rho) \otimes |1\rangle\langle 1| + B\rho B^\dagger \otimes |0\rangle\langle 0|, \quad (8.97)$$

with $c := \|T^*(\mathbb{1})\|_\infty^{-1}$ and $B := \sqrt{\mathbb{1} - cT^*(\mathbb{1})}$.

In other words, Thm.8.22 shows that Hilbert's metric provides a necessary and sufficient condition for the existence of a probabilistic quantum operation which maps $\rho_i \mapsto \rho'_i$ upon success.

Asymptotic convergence and ergodic theory. In this paragraph we consider quantum channels which can be applied repeatedly, i.e., $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$, and analyze the behavior of T^n as $n \in \mathbb{N}$ goes to infinity. Naturally, some of the spectral properties investigated in Chapter 6 will enter the discussion. For instance, depending on whether or not T has eigenvalues of modulus one which are different from one, T^n will or will not converge. In any case the image of T^n will converge to $\mathcal{X}_T$, the space spanned by all eigenvectors corresponding to eigenvalues of modulus one as defined in Eq.(6.65). Similarly, according to Prop.6.3 there will be an ascending subsequence $n_i \in \mathbb{N}$ such that $T^{n_i} \rightarrow T_\phi$ will converge to T_ϕ which is a projection onto χ_T . In other words

$$\|T^n - T_\phi\| \rightarrow 0 \quad (8.98)$$

where $T_\phi := T_\phi T$ describes the “asymptotic dynamics” which is studied in greater detail in Thm.6.16. Recall from Chapter 6 that T_ϕ is defined in terms of the spectral decomposition of T where all eigenvalues of modulus smaller than one are set to zero while the peripheral eigensystem is kept unchanged. Similarly, T_ϕ is nothing but the projection corresponding to the peripheral eigenvectors.

The question which is addressed in the following is: how fast is the convergence in Eq.(8.98) ?

Clearly, some of the previous results on contractivity may be applied. For instance Thm.8.17 implies that

$$\|T^n(\rho_1) - T^n(\rho_2)\|_1 \leq (1 - \epsilon)^n \|\rho_1 - \rho_2\|_1,$$

but it leaves us with having to determine $(1 - \epsilon)$.

A clearer picture about the asymptotic behavior¹³ can be obtained by having a closer look at spectral properties: recall that the spectral radius of a positive and trace-preserving map is one, so that $(T - T_\phi)$ has spectral radius equal to μ where $\mu := \sup_{\lambda \in \text{spec}(T)} \{|\lambda| < 1\}$ is the largest modulus of the eigenvalues of T which are in the interior of the unit disc. Note further that

$$T^n - T_\phi^n = (T - T_\phi)^n, \quad (8.99)$$

¹³which might, however, lead to quite ridiculous bounds for finite n

since $T_\varphi T = TT_\varphi = T_\varphi^2$. Taken together these facts suggest that μ governs the asymptotic behavior and we will see indeed that

$$\|T^n - T_\varphi^n\| = \mathcal{O}(\nu^n), \quad \text{for all } \nu > \mu. \quad (8.100)$$

Before we proceed, some remarks are in order: (i) note that the choice of the norm is only relevant when it comes to decorating Eq.(8.100) with actual constants; most of the time we will consider $\|\hat{T}^n - \hat{T}_\varphi^n\|_\infty$, (ii) if T has only one eigenvalue of modulus one with corresponding fixed point density matrix ρ_∞ , then $T_\varphi(\cdot) = \text{tr}[\cdot]\rho_\infty$ and $T_\varphi = T_\phi = T^\infty$, (iii) it is not a typo that we do not have $\nu \geq \mu$ in Eq.(8.100). In order to understand the latter, consider the following example:

Example 8.1 (Slow convergence of non-diagonalizable channels) *A simple example of a channel whose convergence is slower than $\mathcal{O}(\mu^n)$ is a classical channel with $0 < \mu < 1$, characterized by a stochastic matrix $S \in \mathcal{M}_3(\mathbb{R}_+)$ of the form*

$$S = \begin{pmatrix} \mu & 0 & 0 \\ 1-\mu & \mu & 0 \\ 0 & 1-\mu & 1 \end{pmatrix}, \quad \text{leading to } S^\infty = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}. \quad (8.101)$$

In this case we can compute $S^n - S^\infty$ explicitly and verify that $\|S^n - S^\infty\|$ is asymptotically bounded by $\mu^n n$ both, from below and from above. A general proof of such an asymptotic behavior is given Thm.8.23 where it is traced back to the appearance of non-trivial Jordan blocks.

Note that this example can be translated to the context of quantum channels by defining a quantum channel on $\mathcal{M}_3(\mathbb{C})$ as $T(\cdot) := \sum_{i,j} S_{i,j} \langle j | \cdot | j \rangle | i \rangle \langle i |$ which then displays the same type of asymptotic behavior since $\hat{T} = S \oplus 0$. The main purpose of this example is to show that quantum channels with large Jordan blocks exist and can be easily constructed: extending the example to $\mathcal{M}_d(\mathbb{C})$ leads to a Jordan block of size $d - 1$.

The above example suggests that the asymptotic behavior is governed by μ and the size (denote it by d_μ) of the largest Jordan block corresponding to μ . In the following we will see that there are indeed n -independent constants C_1, C_2 such that

$$C_1 \mu^n n^{d_\mu-1} \leq \|T^n - T_\varphi^n\| \leq C_2 \mu^n n^{d_\mu-1}. \quad (8.102)$$

Proving this requires a Lemma:

Lemma 8.5 *Consider an upper-triangular matrix $X \in \mathcal{M}_D(\mathbb{C})$ with decomposition $X = \Lambda + N$ into a diagonal part Λ and a strictly upper-diagonal part N (i.e., $N_{i,j} = 0$ if $i \geq j$). If $\|\cdot\|$ is a sub-multiplicative norm, $\|\Lambda\| \leq 1$ and $n \in \mathbb{N}$, then*

$$\|X^n\| \leq \|\Lambda^n\| + C_{D,n} \|\Lambda\|^{n-D+1} \max\{\|N\|, \|N\|^{D-1}\}, \quad (8.103)$$

with $C_{D,n} = (D-1)n^{D-1}$. If in addition $2(D-1) \leq n$ then Eq.(8.103) holds with $C_{D,n} = (D-1)\binom{n}{D-1}$.

Similarly, if X is a Jordan block with diagonal part $\Lambda = \lambda \mathbf{1}$, then for all $n \in \mathbb{N}$ and natural numbers $k_0 \leq \min\{n, D-1\}$

$$|\lambda|^{n-k_0} \binom{n}{k_0} \leq \|X^n\|_\infty \leq \sum_{k=0}^{\min\{n, D-1\}} |\lambda|^{n-k} \binom{n}{k}. \quad (8.104)$$

PROOF Looking at the binomial expansion of $(\Lambda + N)^n$ we first observe that all monomials in which N appears more than $D-1$ times will vanish since they are products of more than $D-1$ strictly upper-triangular matrices. Hence, by using the basic properties of the norm we obtain

$$\|X^n\| \leq \|\Lambda^n\| + \sum_{k=1}^{\min\{n, D-1\}} \binom{n}{k} \|N\|^k \|\Lambda\|^{n-k}, \quad (8.105)$$

which leads to Eq.(8.103) when bounding either $\binom{n}{k} \leq n^{D-1}$ or $\binom{n}{k} \leq \binom{n}{D-1}$ if $n \geq 2(D-1)$ and then choosing k extremal in each of the three factors.

Now assume that X is a Jordan block. Since then $[\Lambda, N] = 0$ we have $X^n = \sum_k \lambda^{n-k} N^k \binom{n}{k}$, where k again runs from 0 to $D-1$. X^n is thus a Topelitz matrix whose first row has entries of the form $\lambda^{n-k} \binom{n}{k}$. Since the largest singular value, i.e. the operator norm, is bounded from below by the largest modulus of all entries, we obtain the l.h.s. of Eq.(8.104). The r.h.s. is obtained by using the triangle inequality for the norm together with $\|N\|_\infty = 1$. $\square$

Using this Lemma we will now derive two related results about the asymptotic behavior of T^n —based on either Jordan decomposition or Schur decomposition of T . While the first one allows us to pinpoint the exact asymptotic scaling, the second one enables us to provide explicit (albeit generally pessimistic) constants which depend only on n , μ and the underlying Hilbert space dimension.

For the following consider the Jordan decomposition $\hat{T} = A(\bigoplus_{k=1}^K J_k(\lambda_k))A^{-1}$ where k labels the different eigenvalues λ_k of T according to their geometric multiplicity. That is, K is the number of eigenvectors. We define the *Jordan condition number* as $\kappa_T := \inf_A \|A\|_\infty \|A^{-1}\|_\infty$ where the infimum is taken over all similarity transformation A which bring $\hat{T}$ to Jordan normal form.

Theorem 8.23 (Asymptotic convergence I) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a positive and trace preserving linear map. Denote by $\mu := \sup_{\lambda \in \text{spec}(T)} \{|\lambda| < 1\}$ the largest modulus of all eigenvalues which are in the interior of the unit disc, and by d_μ the dimension of the largest Jordan block corresponding to an eigenvalue of modulus μ . Then there exists a constant $C_1 > 0$ such that for all $n \in \mathbb{N}$*

$$C_1 \mu^n n^{d_\mu-1} \leq \|\hat{T}^n - \hat{T}_\varphi^n\|_\infty. \quad (8.106)$$

Moreover, this inequality holds for $C_1 = \begin{cases} \kappa_T^{-1} & \text{if } d_\mu = 1, \\ \kappa_T^{-1} (\mu(d_\mu - 1))^{1-d_\mu} & \text{if } d_\mu \leq n + 1. \end{cases}$

Similarly, there exists a $C_2 \in \mathbb{R}$ such that for all $n \in \mathbb{N}$ we have

$$\|\hat{T}^n - \hat{T}_\varphi^n\|_\infty \leq C_2 \mu^n n^{d_\mu-1}. \quad (8.107)$$

If $\hat{T}$ is diagonalizable (i.e., all Jordan blocks are one-dimensional), we can choose $C_2 = \kappa_T$.

PROOF By definition of T_φ we have that $\hat{T} - \hat{T}_\varphi = A(\bigoplus_{k:|\lambda_k|<1} J_k(\lambda_k))A^{-1}$ (assuming, with some abuse of notation, that missing Jordan blocks are properly replaced by blocks of zero matrices). Exploiting Eq.(8.99) together with submultiplicativity of the norm we obtain

$$\kappa_T^{-1} \|J^n\|_\infty \leq \|\hat{T}^n - \hat{T}_\varphi^n\|_\infty \leq \kappa_T \|J^n\|_\infty \quad \text{with } J := \bigoplus_{k:|\lambda_k|<1} J_k(\lambda_k). \quad (8.108)$$

It follows from Eq.(8.104) that for every pair of eigenvalues with $|\lambda_i| < |\lambda_j|$ there is an $n_0 \in \mathbb{N}$ such that $\|J_i(\lambda_i)^n\|_\infty < \|J_j(\lambda_j)^n\|_\infty$ holds for the norms of the corresponding Jordan blocks for all $n \geq n_0$. Moreover, if two eigenvalues have equal magnitude, then the Jordan block with the larger dimension has the larger norm.¹⁴ Hence, there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ we have

$$\|J^n\|_\infty \leq \|X^n\|_\infty$$

where X is chosen to be the largest Jordan block corresponding to an eigenvalue of modulus μ . At this point the r.h.s. of Eq.(8.104) becomes applicable with $D = d_\mu$ and we arrive at the desired result in Eq.(8.107) by using that $\binom{n}{k} \leq n^{d_\mu-1}$ is implied by $k \leq d_\mu - 1$. Note that we can drop the requirement that $n \geq n_0$ by appropriately rescaling C_2 .

The lower bound stated in Eq.(8.106) is similarly obtained by using $\|J^n\|_\infty \geq \|X^n\|_\infty$ and inserting the l.h.s. of Eq.(8.108) together with Eq.(8.104) after setting $k_0 = d_\mu - 1$ and bounding $\binom{n}{k_0} \geq (n/k_0)^{k_0}$. $\square$

Note that if $\hat{T}$ is diagonalizable (as it holds for almost all channels), then the Jordan decomposition used in Thm.8.23 yields

$$\kappa_T^{-1} \mu^n \leq \|\hat{T}^n - \hat{T}_\varphi^n\|_\infty \leq \kappa_T \mu^n \quad (8.109)$$

In some cases the Jordan condition number κ_T can be bounded in terms of the condition number $\kappa(\sigma) := \|\sigma\|_\infty \|\sigma^{-1}\|_\infty$ of a fixed point density matrix $\sigma = T(\sigma)$:

Proposition 8.5 (Jordan condition number and detailed balance) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a Hermiticity preserving linear map and $\Sigma : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ a linear map such that $\hat{\Sigma} > 0$ is a positive definite matrix. If*

$$\Sigma T^* = T \Sigma, \quad (8.110)$$

then the Jordan condition number satisfies $\kappa_T \leq \sqrt{\kappa(\hat{\Sigma})}$. In particular, if $\Sigma(X) = \sqrt{\sigma} X \sqrt{\sigma}$ for some positive definite $\sigma \in \mathcal{M}_d(\mathbb{C})$ (which is a fixed point of T if $T^(\mathbb{1}) = \mathbb{1}$), then $\kappa_T \leq \sqrt{\kappa(\sigma)}$.*

¹⁴This can be seen by first noting that for Jordan blocks of equal size $\|J(\lambda)^n\|_\infty = \|J(|\lambda|)^n\|_\infty$ and then using that $J(|\lambda|)^n$ contains any smaller Jordan block with eigenvalue $|\lambda|$ as principal submatrix whose operator norm thus forms a lower bound.

PROOF Expressed in terms of the matrix representation $\hat{T}$ the “detailed balance” type condition of Eq.(8.110) reads $\hat{\Sigma}\hat{T}^\dagger = \hat{T}\hat{\Sigma}$. Since $\hat{\Sigma} > 0$ this is equivalent to $\hat{\Sigma}^{1/2}\hat{T}^\dagger\hat{\Sigma}^{-1/2} = \hat{\Sigma}^{-1/2}\hat{T}\hat{\Sigma}^{1/2}$ which is thus a Hermitian matrix so that $\hat{T}$ can be diagonalized by a similarity transformation $\hat{\Sigma}^{1/2}U$ for some unitary U . Since multiplying with the latter doesn’t change the operator norm, we obtain $\kappa_T \leq \kappa(\hat{\Sigma}^{1/2}) = \sqrt{\kappa(\hat{\Sigma})}$ which equals $\sqrt{\kappa(\sigma)}$ if $\hat{\Sigma} = \sigma^{1/2} \otimes \bar{\sigma}^{1/2}$. $\square$

For the general non-diagonalizable case, the constant C_2 (constant w.r.t. n) in Thm.8.23 is somewhat elusive so that Eq.(8.107) becomes a rather weak statement. A look at the proof reveals that we can obtain an explicit C_2 by either making it dependent on the separation of the moduli of eigenvalues, or by replacing d_μ in Eq.(8.107) by the dimension of the largest Jordan block corresponding to *any* eigenvalue in the interior of the unit disk. In the latter case C_2 would still involve the Jordan condition number κ_T which in turn may be bounded in terms of the separation of eigenvalues (see [?]). The subsequent theorem will follow an alternative route which, to some extent, allows to circumvent such issues. It is based on the Schur decomposition $(\hat{T} - \hat{T}_\varphi) = U(\Lambda + N)U^\dagger$ where U is unitary, Λ is diagonal and N is strictly upper triangular.

Theorem 8.24 (Asymptotic convergence II) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a positive and trace preserving linear map. Denote by $\mu := \sup_{\lambda \in \text{spec}(T)} \{|\lambda| < 1\}$ the largest modulus of all eigenvalues which are in the interior of the unit disc. Then for any $n \in \mathbb{N}$*

$$\|\hat{T}^n - \hat{T}_\varphi^n\|_\infty \leq \mu^n + C_{d,n} \mu^{n-d^2+1} \max\{\|N\|_\infty, \|N\|_\infty^{d^2-1}\}, \quad (8.111)$$

with $C_{d,n} = (d^2 - 1)n^{d^2-1}$. If in addition $2(d^2 - 1) \leq n$ then Eq.(8.111) holds with $C_{d,n} = (d^2 - 1)\binom{n}{d^2-1}$. Moreover, we can bound $\|N\|_\infty \leq (\mu + 2\sqrt{d})$.

PROOF Eq.(8.111) is an immediate consequence of Eq.(8.103) together with the spectral properties of T . The operator norm bound for N follows from $\|N\|_\infty \leq \|\Lambda\|_\infty + \|\hat{T}\|_\infty + \|\hat{T}_\varphi\|_\infty$ since $\|\Lambda\|_\infty = \mu$ and the singular values of positive, trace-preserving linear maps on $\mathcal{M}_d(\mathbb{C})$ are bounded by $\sqrt{d}$. $\square$

We will now relate the operator norm convergence of maps, which we have discussed in the last paragraphs, to the trace norm convergence of states. Recall that $\mathcal{X}_T := \text{span}\{X \in \mathcal{M}_d(\mathbb{C}) | \exists \varphi \in \mathbb{R} : T(X) = e^{i\varphi} X\}$ is the complex linear span of all eigenvectors of T with corresponding eigenvalue of modulus one. According to Prop.6.12 $\mathcal{X}_T$ admits a basis consisting of density operators and satisfies $\mathcal{X}_T = T(\mathcal{X}_T) = T_\phi(\mathcal{M}_d(\mathbb{C}))$.

Proposition 8.6 (Convergence towards asymptotic states) *Let $T : \mathcal{M}_d(\mathbb{C}) \rightarrow \mathcal{M}_d(\mathbb{C})$ be a positive and trace preserving linear map and denote by $\Delta_T(\rho) := \|\rho - T_\phi(\rho)\|_1$ the trace-norm distance between $\rho \in \mathcal{M}_d(\mathbb{C})$ and its projection onto $\mathcal{X}_T$. Then for any density operator $\rho \in \mathcal{M}_d(\mathbb{C})$ and $n \in \mathbb{N}$*

$$\Delta_T(T^n(\rho)) \leq \sqrt{d/2} \|\hat{T}^n - \hat{T}_\varphi^n\|_\infty \Delta_T(\rho). \quad (8.112)$$

Conversely, if the supremum is taken over all density operators $\rho \in \mathcal{M}_d(C)$, then

$$\sup_{\rho} \Delta_T(T^n(\rho)) \geq \frac{1}{4\sqrt{d}} \|\hat{T}^n - \hat{T}_{\varphi}^n\|_{\infty}. \quad (8.113)$$

PROOF First note that since $T_{\phi}T = TT_{\phi} = T_{\varphi} = T_{\varphi}T_{\phi}$ we have $\Delta_T(T^n(\rho)) = \|T^n(\rho - T_{\phi}(\rho))\|_1 = \|(T^n - T_{\varphi}^n)(\rho - T_{\phi}(\rho))\|_1$. So for any density operator ρ we obtain

$$\frac{\Delta_T(T^n(\rho))}{\Delta_T(\rho)} = \frac{\|(T^n - T_{\varphi}^n)(\rho - T_{\phi}(\rho))\|_1}{\|\rho - T_{\phi}(\rho)\|_1} \quad (8.114)$$

$$\leq \frac{\sqrt{d}}{2} \sup_{\psi \perp \phi} \|(T^n - T_{\varphi}^n)(|\psi\rangle\langle\psi| - |\phi\rangle\langle\phi|)\|_2 \quad (8.115)$$

$$\leq \sqrt{d/2} \sup_{X \in \mathcal{M}_d(\mathbb{C})} \frac{\|(T^n - T_{\varphi}^n)(X)\|_2}{\|X\|_2} \quad (8.116)$$

$$= \sqrt{d/2} \|(T^n - T_{\varphi}^n)\|_{2 \rightarrow 2}. \quad (8.117)$$

Here the first inequality uses Lemma 8.3 (where the optimization is over all pairs of orthogonal unit vectors in $\mathbb{C}^d$) together with $\|\cdot\|_1 \leq \sqrt{d}\|\cdot\|_2$ from Eq.(8.7). The second inequality uses that $\| |\psi\rangle\langle\psi| - |\phi\rangle\langle\phi| \|_2 = \sqrt{2}$. Since $\|(T^n - T_{\varphi}^n)\|_{2 \rightarrow 2} = \|(\hat{T}^n - \hat{T}_{\varphi}^n)\|_{\infty}$ we finally arrive at the inequality in Eq.(8.112).

For the converse statement recall that $\|(T^n - T_{\varphi}^n)\|_{2 \rightarrow 2} = \sup_{X: \|X\|_2 \leq 1} \|(T^n - T_{\varphi}^n)(X)\|_2$. Any such X can be decomposed into four positive parts so that $X = \sum_{i=1}^4 c_i P_i$ with $|c_i| = 1$ and $P_i \geq 0$, $\|P_i\|_2 \leq 1$. Together with $\|\cdot\|_2 \leq \|\cdot\|_1$ this implies

$$\frac{1}{4} \|(T^n - T_{\varphi}^n)\|_{2 \rightarrow 2} \leq \sup_{P \geq 0} \{ \|(T^n - T_{\varphi}^n)(P)\|_1 \mid \|P\|_2 \leq 1 \}.$$

Eq.(8.113) then follows by observing that $\|P\|_2 \leq 1$ implies $\text{tr}[P] \leq \sqrt{d}$. $\square$

8.8 Continuity bounds